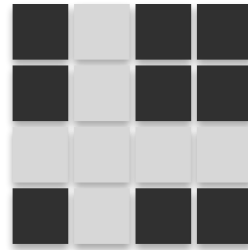


Week 03 · Diffusion and generated media

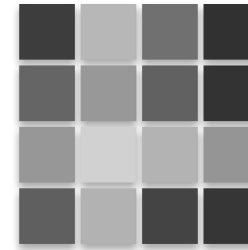
OPEN / WEEK 03

The consent-safe poster

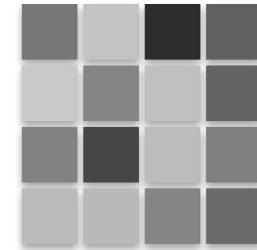
A team requests a synthetic poster. The output looks vivid but may show a recognisable person. How do generation and review interact?



clean



less noise



more noise

Generation uses learned reverse steps

First vote: what would you do?

QUESTION / WEEK 03

A team requests a synthetic poster. The output looks vivid but may show a recognisable person. How do generation and review interact?

Choose an initial answer and name one piece of evidence you need.

VOTE FIRST • GIVE A REASON

A Accept the first plausible result

B Check the source or measurement

Which source or test could settle this?

The journey through today's case

OPEN / WEEK 03

We will predict, connect an analogy, trace a mechanism, work an example, then explain a decision.

1

Forward noising

2

Denoising model

3

Conditioning

4

Image review

1. Predict · Forward noising

QUESTION / WEEK 03

Which direction is used while preparing training pairs?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A Forward: progressively add noise to a clean example.

B The model learns by removing noise before any noise is added.

Which source or test could settle this?

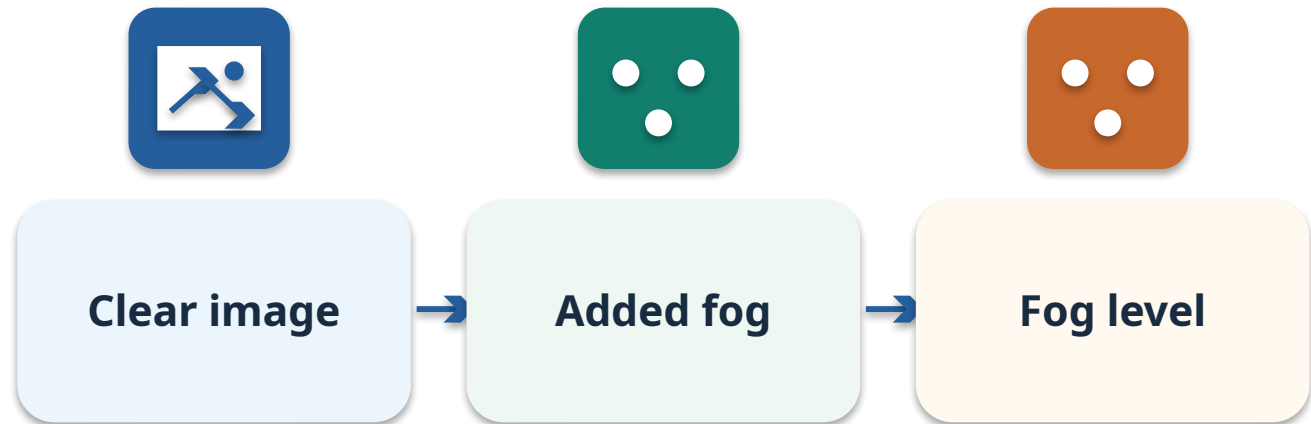
Forward noising · a familiar picture

ANALOGY / WEEK 03

A photograph becomes harder to see as fog builds in small steps.

What matches our case, and where does the analogy stop?

A photograph becomes harder to see as fog builds in small steps.



What matches the technical process?

Forward noising · map the parts

MAPPING / WEEK 03

Match each everyday part to the technical part.

Analogy limit: Noise is mathematically sampled; real fog is not the model's training process.

FAMILIAR SCENE

Clear image

Added fog

Fog level

TECHNICAL IDEA

training example

forward noise

time step

The analogy has a limit. Read it on the left.

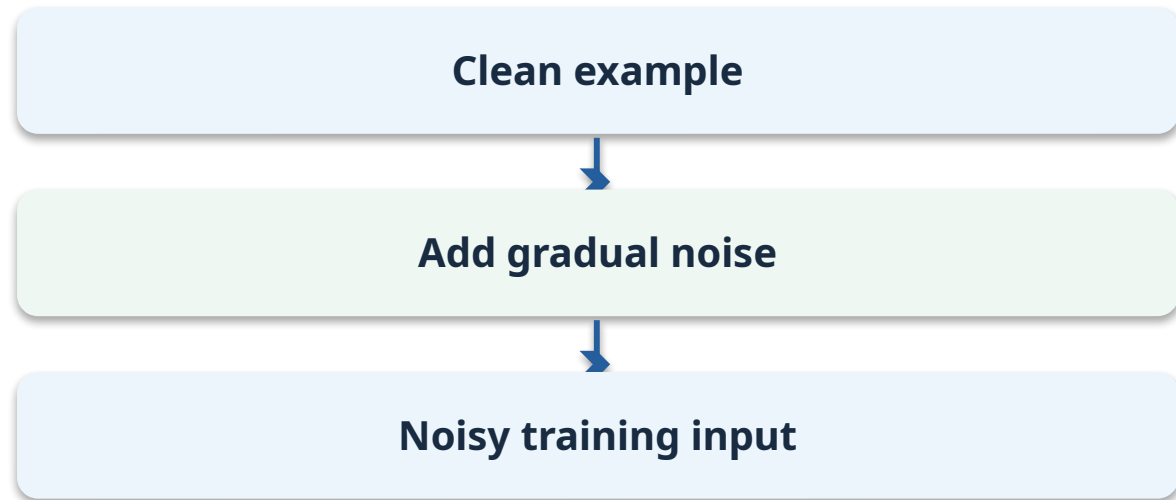
Forward noising · how it works

MECHANISM / WEEK 03

In a common training setup, progressively add noise to a clean image.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Forward noising · work one case

WORKED / WEEK 03

At low noise the L-shaped grid remains visible; at high noise most cells are obscured.

Point to the evidence before giving your conclusion.

GIVEN

At low noise the L-shaped grid remains visible; at high noise most cells are obscured.

QUESTION

Which direction is used while preparing training pairs?

SUPPORTED ANSWER

Forward: progressively add noise to a clean example.

Forward noising · answer and explain

REVEAL / WEEK 03

Forward: progressively add noise to a clean example.

Tempting mistake: The model learns by removing noise before any noise is added.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

The model learns by removing noise before any noise is added.

SUPPORTED EXPLANATION

Forward: progressively add noise to a clean example.

Noise is mathematically sampled; real fog is not the model's training process.

2. Predict · Denoising model

QUESTION / WEEK 03

What extra information makes the toy restoration easier?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A It has access to the original clean grid.

B The toy denoiser proves real model image quality.

Which source or test could settle this?

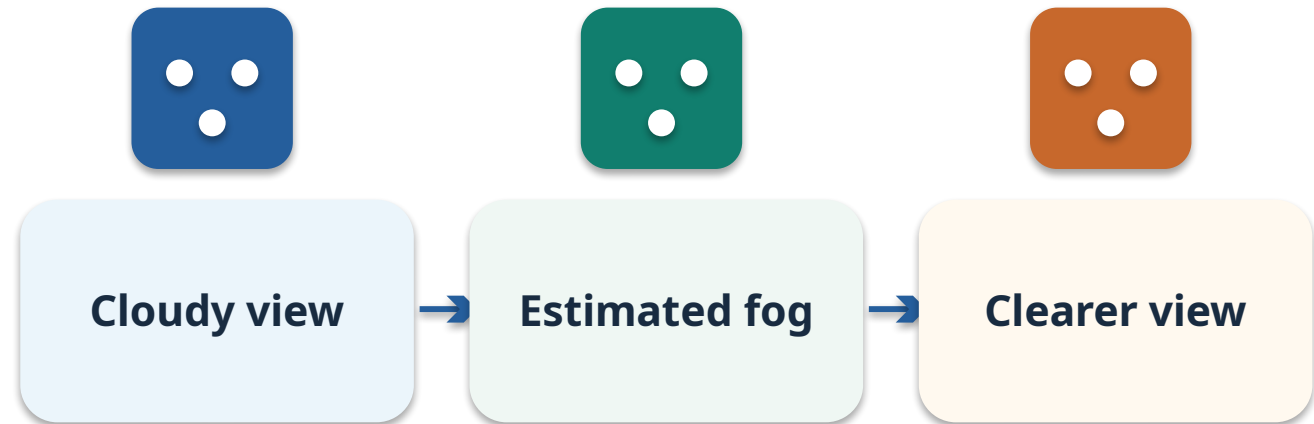
Denoising model · a familiar picture

ANALOGY / WEEK 03

A restorer removes a little fog at a time after learning from many examples.

What matches our case, and where does the analogy stop?

A restorer removes a little fog at a time after learning from many examples.



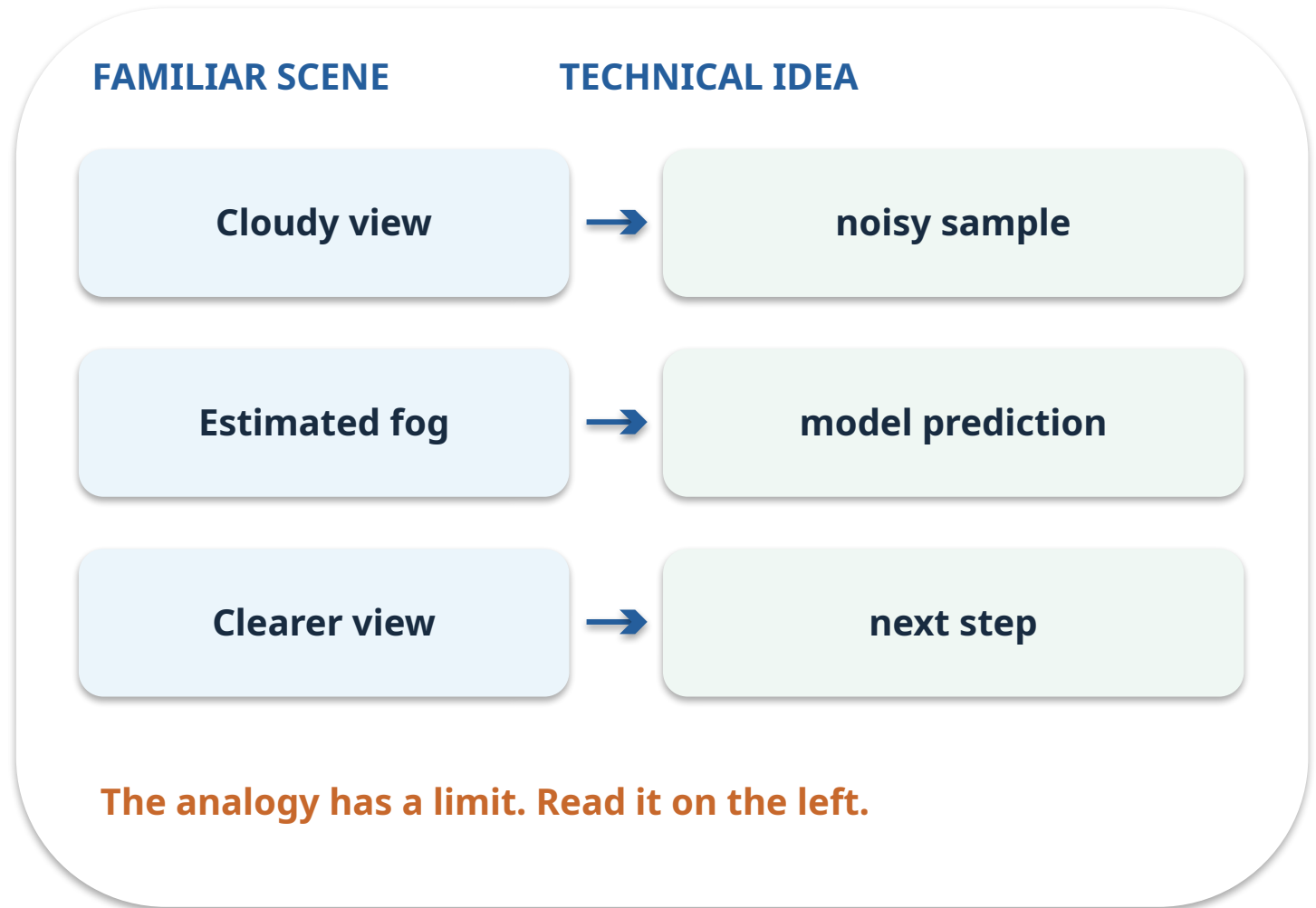
What matches the technical process?

Denoising model · map the parts

MAPPING / WEEK 03

Match each everyday part to the technical part.

Analogy limit: A real model does not secretly have the exact clean target at generation time.



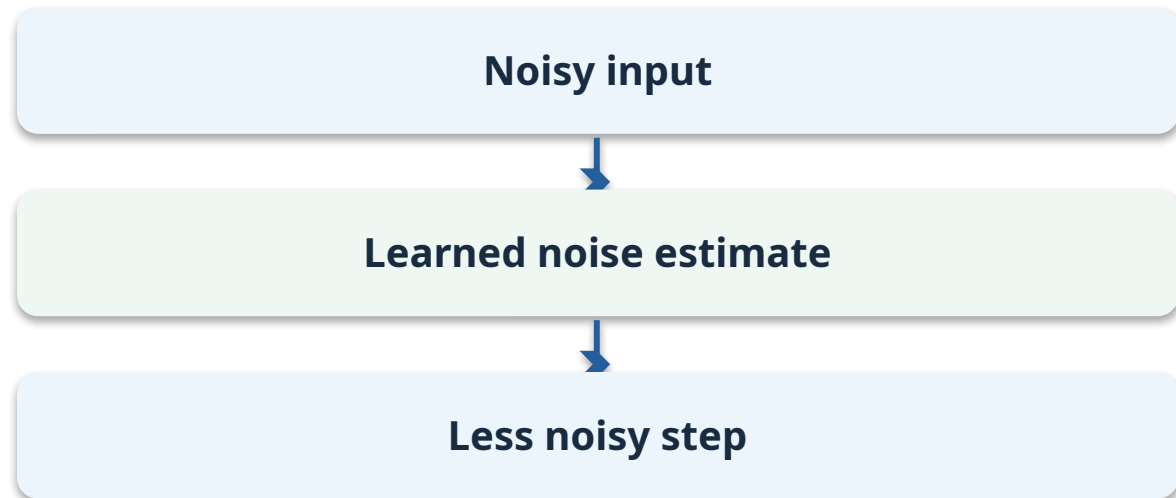
Denoising model · how it works

MECHANISM / WEEK 03

A learned model estimates how to remove noise at each step; the course simulation uses the known clean target.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Denoising model · work one case

WORKED / WEEK 03

The prepared notebook's toy restorer uses a clean target to illustrate steps; label that shortcut.

Point to the evidence before giving your conclusion.

GIVEN

The prepared notebook's toy restorer uses a clean target to illustrate steps; label that shortcut.

QUESTION

What extra information makes the toy restoration easier?

SUPPORTED ANSWER

It has access to the original clean grid.

Denoising model · answer and explain

REVEAL / WEEK 03

It has access to the original clean grid.

Tempting mistake: The toy denoiser proves real model image quality.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

The toy denoiser proves real model image quality.

SUPPORTED EXPLANATION

It has access to the original clean grid.

A real model does not secretly have the exact clean target at generation time.

3. Predict · Conditioning

QUESTION / WEEK 03

What stays fixed in a fair output comparison?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A Keep the brief and checklist fixed; record changed generation settings.

B A detailed prompt guarantees exact text in images.

Which source or test could settle this?

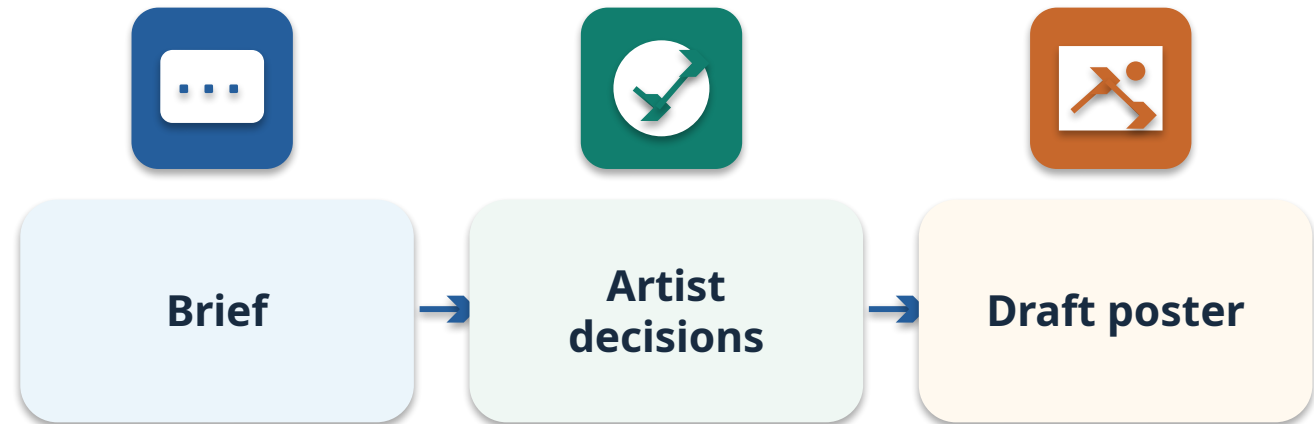
Conditioning · a familiar picture

ANALOGY / WEEK 03

A poster brief steers an artist, but leaves room for choices.

What matches our case, and where does the analogy stop?

A poster brief steers an artist, but leaves room for choices.



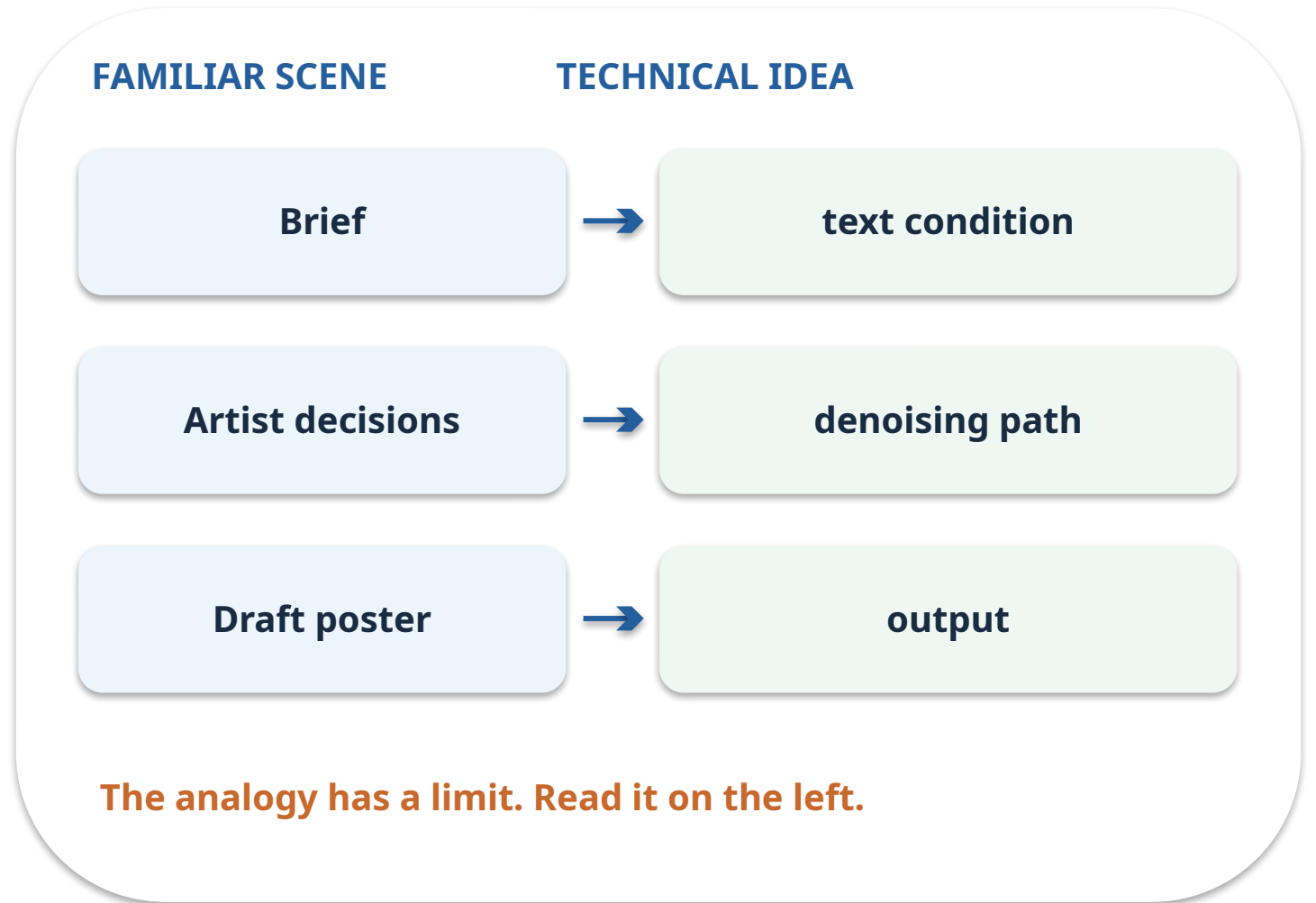
What matches the technical process?

Conditioning · map the parts

MAPPING / WEEK 03

Match each everyday part to the technical part.

Analogy limit: The brief does not guarantee that every requested detail appears correctly.



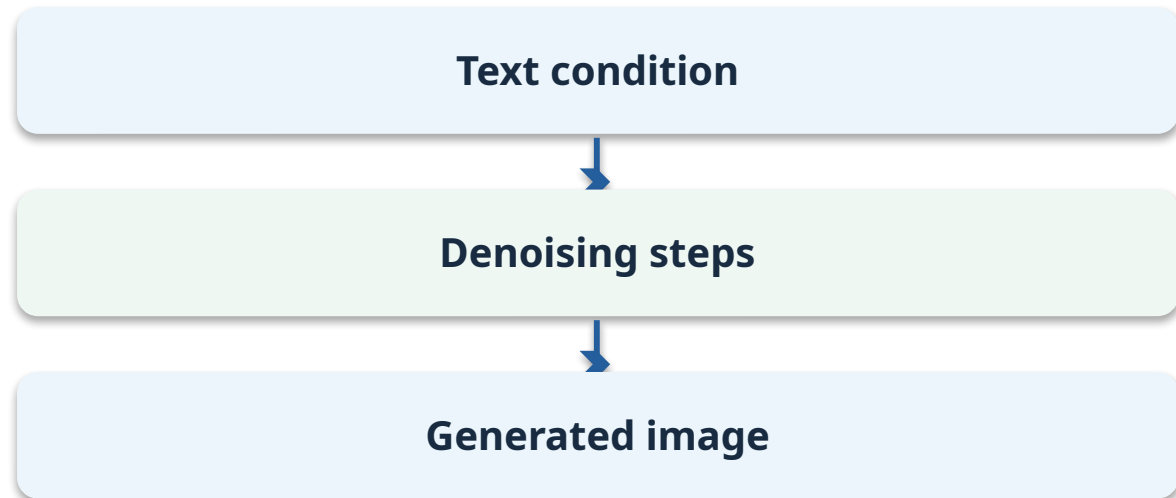
Conditioning · how it works

MECHANISM / WEEK 03

Text or other inputs can guide generation. The condition does not guarantee accurate details.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Conditioning · work one case

WORKED / WEEK 03

Use “blue campus poster” for both trials; compare date lettering and layout separately.

Point to the evidence before giving your conclusion.

GIVEN

Use “blue campus poster” for both trials; compare date lettering and layout separately.

QUESTION

What stays fixed in a fair output comparison?

SUPPORTED ANSWER

Keep the brief and checklist fixed; record changed generation settings.

Conditioning · answer and explain

REVEAL / WEEK 03

Keep the brief and checklist fixed; record changed generation settings.

Tempting mistake: A detailed prompt guarantees exact text in images.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

A detailed prompt guarantees exact text in images.

SUPPORTED EXPLANATION

Keep the brief and checklist fixed; record changed generation settings.

The brief does not guarantee that every requested detail appears correctly.

4. Predict · Image review

QUESTION / WEEK 03

What two reasons justify holding this image?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A Consent is unresolved and the date detail is incorrect.

B A synthetic image needs no rights or quality review.

Which source or test could settle this?

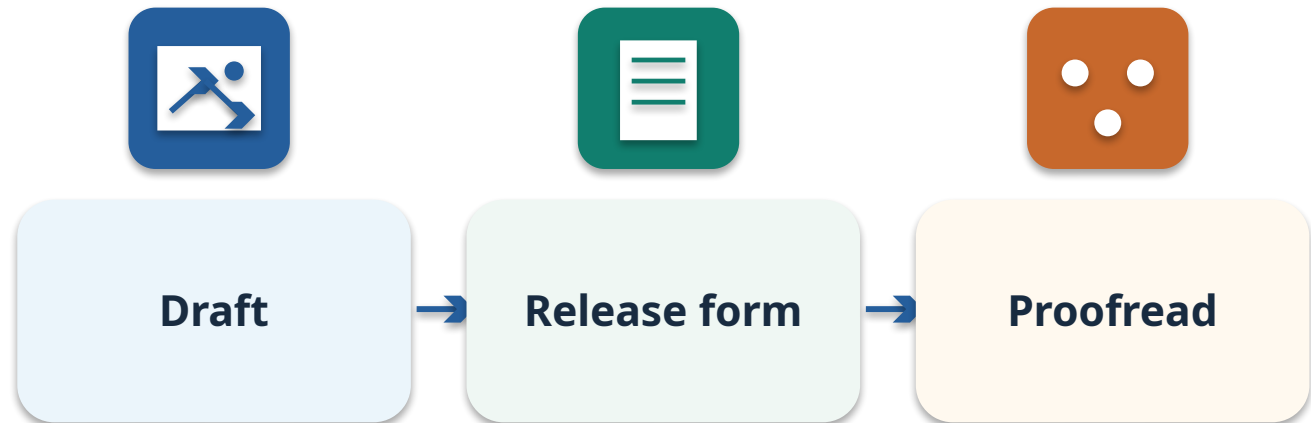
Image review · a familiar picture

ANALOGY / WEEK 03

A print editor checks a draft before it reaches the public.

What matches our case, and where does the analogy stop?

A print editor checks a draft before it reaches the public.



What matches the technical process?

Image review · map the parts

MAPPING / WEEK 03

Match each everyday part to the technical part.

Analogy limit: A release form cannot be inferred from pixels; ask for documented permission.

FAMILIAR SCENE

Draft

Release form

Proofread

TECHNICAL IDEA

generated image

consent check

image detail check

The analogy has a limit. Read it on the left.

Image review · how it works

MECHANISM / WEEK 03

Check whether text, people, permissions and task-specific details meet the brief.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW

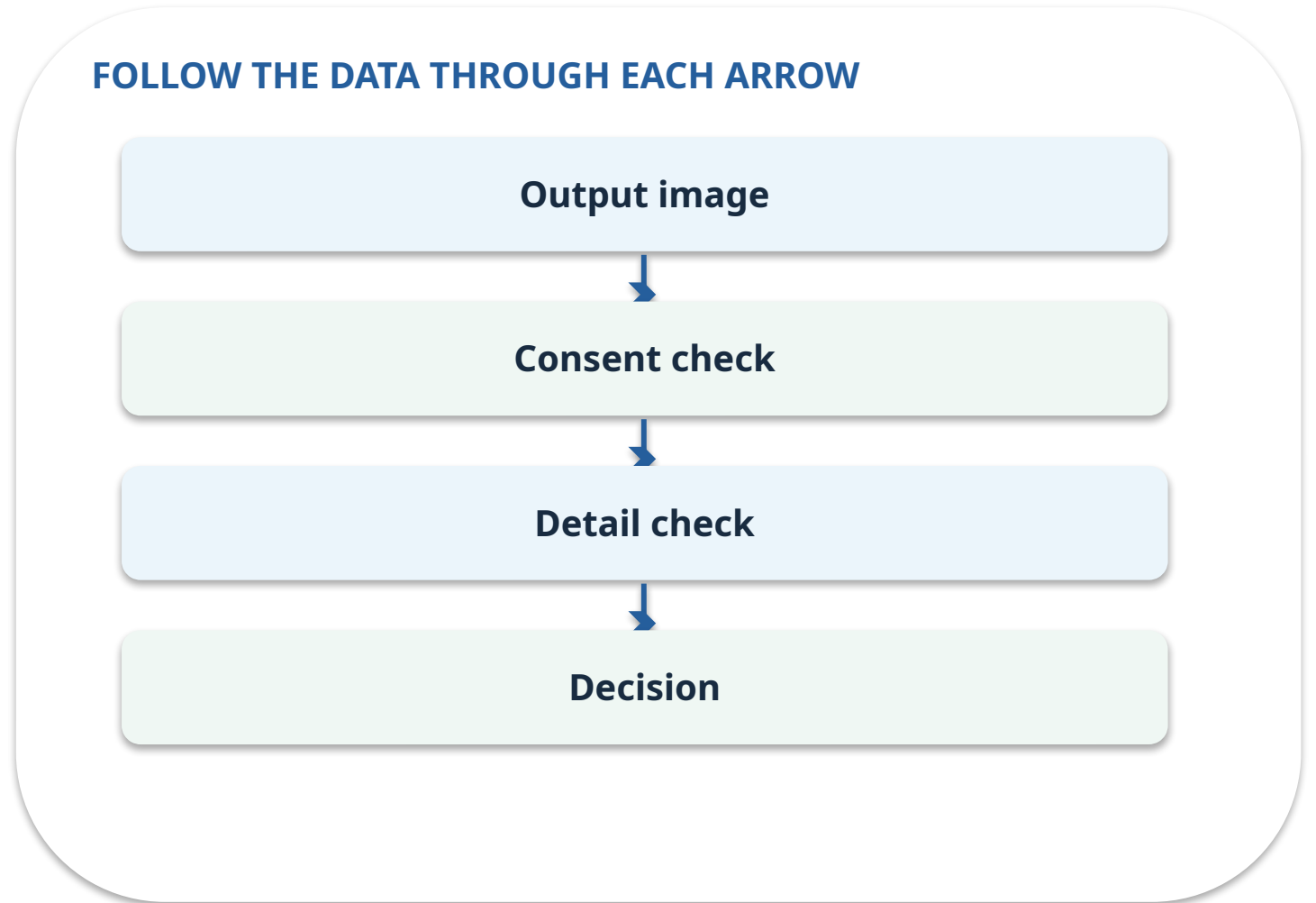


Image review · work one case

WORKED / WEEK 03

The poster is attractive, but a face resembles a classmate and the date is wrong: hold publication.

Point to the evidence before giving your conclusion.

GIVEN

The poster is attractive, but a face resembles a classmate and the date is wrong: hold publication.

QUESTION

What two reasons justify holding this image?

SUPPORTED ANSWER

Consent is unresolved and the date detail is incorrect.

Image review · answer and explain

REVEAL / WEEK 03

Consent is unresolved and the date detail is incorrect.

Tempting mistake: A synthetic image needs no rights or quality review.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

A synthetic image needs no rights or quality review.

SUPPORTED EXPLANATION

Consent is unresolved and the date detail is incorrect.

A release form cannot be inferred from pixels; ask for documented permission.

Evidence trail: input → result

WORKED / WEEK 03

Noise obscures the L at high strength.

A toy restoration with access to the clean target restores it; a real model does not have that target.

GIVEN

Clean grid has an L-shaped bright pattern.

QUESTION

Noise obscures the L at high strength.

SUPPORTED ANSWER

A toy restoration with access to the clean target restores it; a real model does not have that target.

Notebook: run the prepared example

PRACTICE / WEEK 03

Run a denoising illustration and a consent checklist.

Read the original output before editing.

1

Inspect a 4 by 4 synthetic clean image

2

Add reproducible noise at two strengths

3

Use a transparent toy denoising demonstration

4

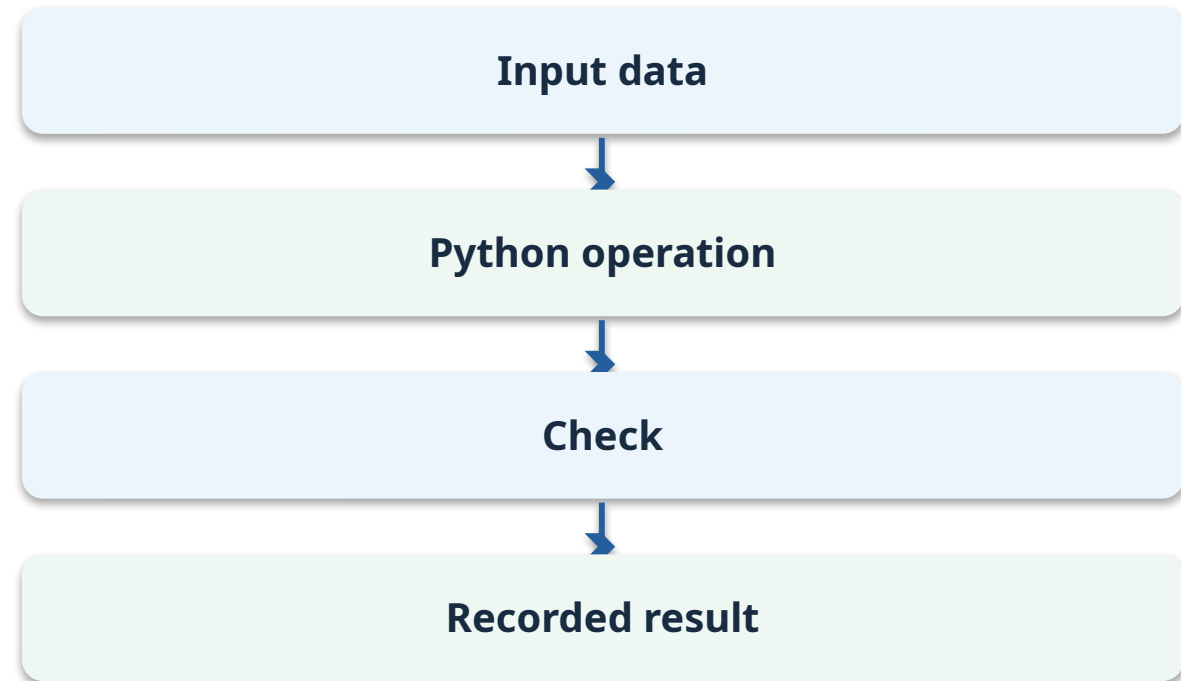
Compare two prepared descriptions using one rubric

Find the input and check in Python

PRACTICE / WEEK 03

Locate the supplied input, the transformation, the checking line and the printed result.

FOLLOW THE DATA THROUGH EACH ARROW



Predict one controlled change

PRACTICE / WEEK 03

Increase noise strength from 0.15 to 0.45 and compare measured error.

What should change in the result? What stays fixed?

Before: original case

After: one change

Keep all other inputs fixed.

Run, compare and record

PRACTICE / WEEK 03

Record the input, the one changed setting, the observed result and a reason for any difference.

Input	Setting	Result
Original	baseline	record
Changed	one edit	compare

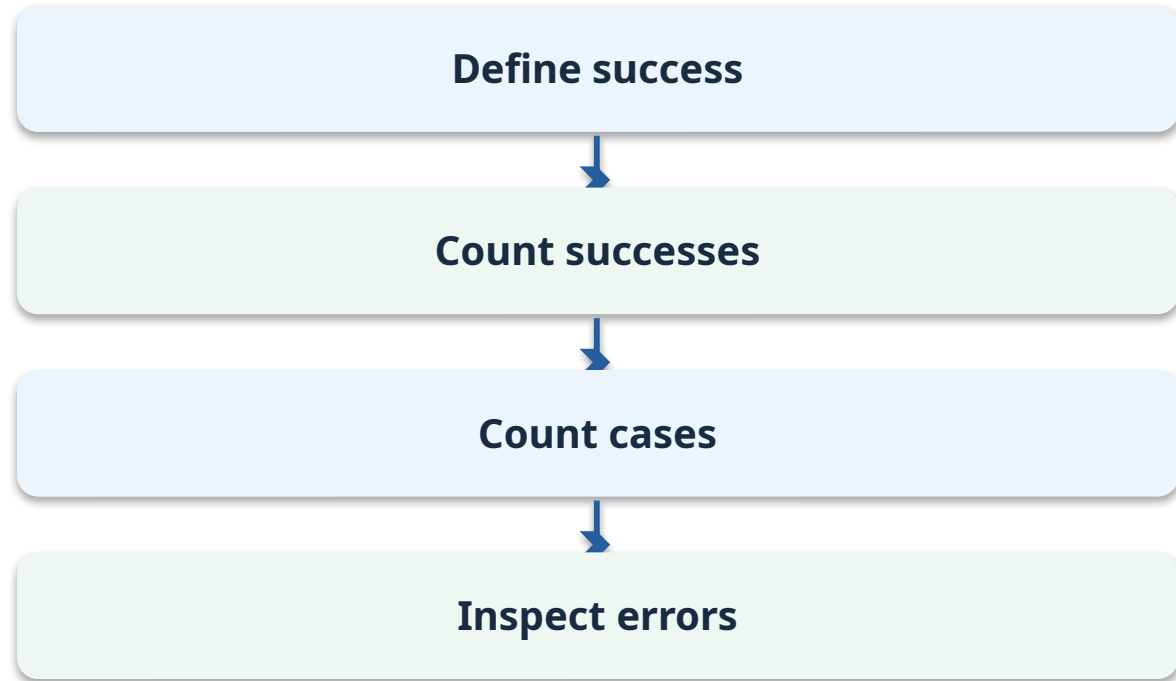
Measure without hiding the denominator

CHECK / WEEK 03

Mean absolute pixel error; plus a human quality checklist

Write the numerator, denominator and one failure case.

FOLLOW THE DATA THROUGH EACH ARROW



Error detective: find the layer

CHECK / WEEK 03

Plausible wrong claim:
Removing random noise with
the original image in hand
reproduces diffusion
generation.

Correct it using the case
evidence.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

**Removing random noise with the original image in hand
reproduces diffusion generation.**

SUPPORTED EXPLANATION

**This exercise visualizes steps only. Actual generation uses
learned denoising without a clean target at inference.**

Name a test that would catch this failure.

Transfer to a new situation

CHECK / WEEK 03

A poster has readable-looking text that spells the event date incorrectly. Which check catches the error?

Explain which source or test would settle it.

VOTE FIRST • GIVE A REASON

A Use the visible evidence

B Flag missing or unsafe evidence

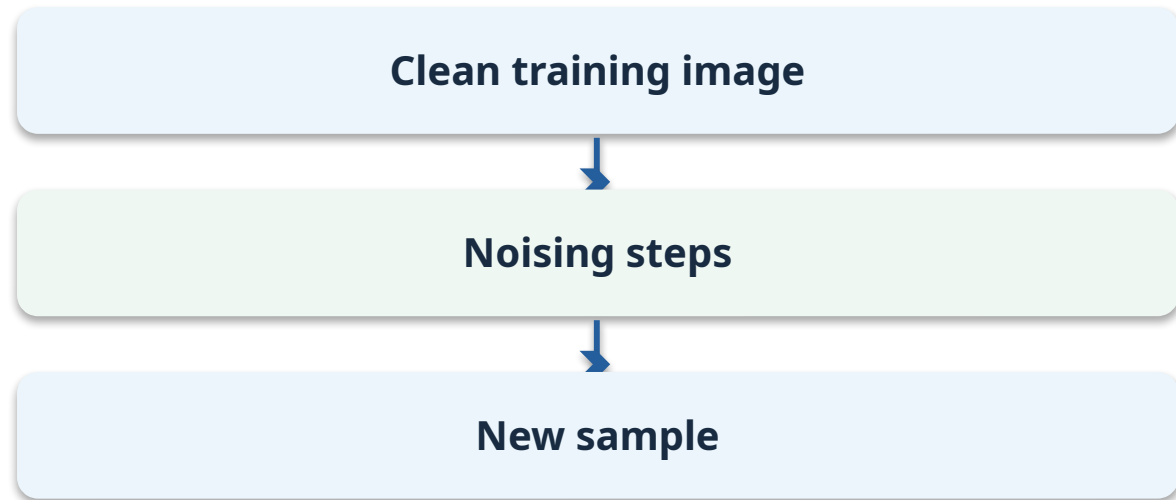
Which source or test could settle this?

Explain it to a partner

CHECK / WEEK 03

Without reading the slide, name the input, one processing step and the check. Your partner should ask “How do you know?”

FOLLOW THE DATA THROUGH EACH ARROW



Exam rehearsal · explain and apply

CHECK / WEEK 03

Distinguish noising during training from denoising during generation.

Write a short answer, then compare with the guide.

1

State the core idea

2

Use a case detail or calculation

3

Add a limitation

4

Compare with the Word guide

Exam rehearsal · explain and apply

CHECK / WEEK 03

Does conditioning guarantee the requested text will be correct?

Write a short answer, then compare with the guide.

1

State the core idea

2

Use a case detail or calculation

3

Add a limitation

4

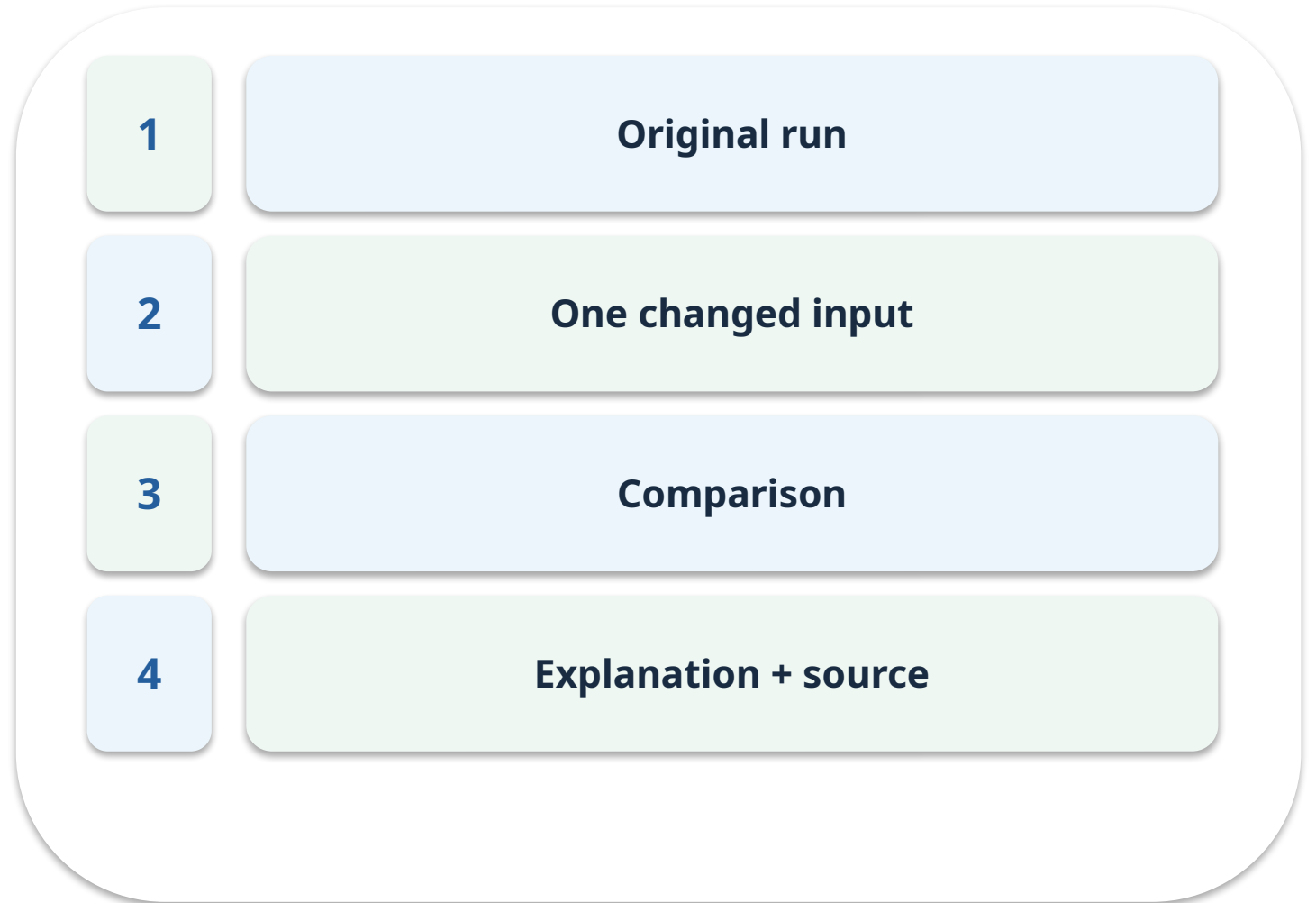
Compare with the Word guide

What to hand in

PRACTICE / WEEK 03

Step comparison and three task-specific image observations.

Include the one changed input and an error explanation.



Three ideas to recall next week

CHECK / WEEK 03

Noise becomes an image through repeated learned denoising; check consent and details.

Read the linked official sources in the Word guide and redraw one diagram from memory.

1

Forward noising

2

Denoising model

3

Conditioning

4

Image review